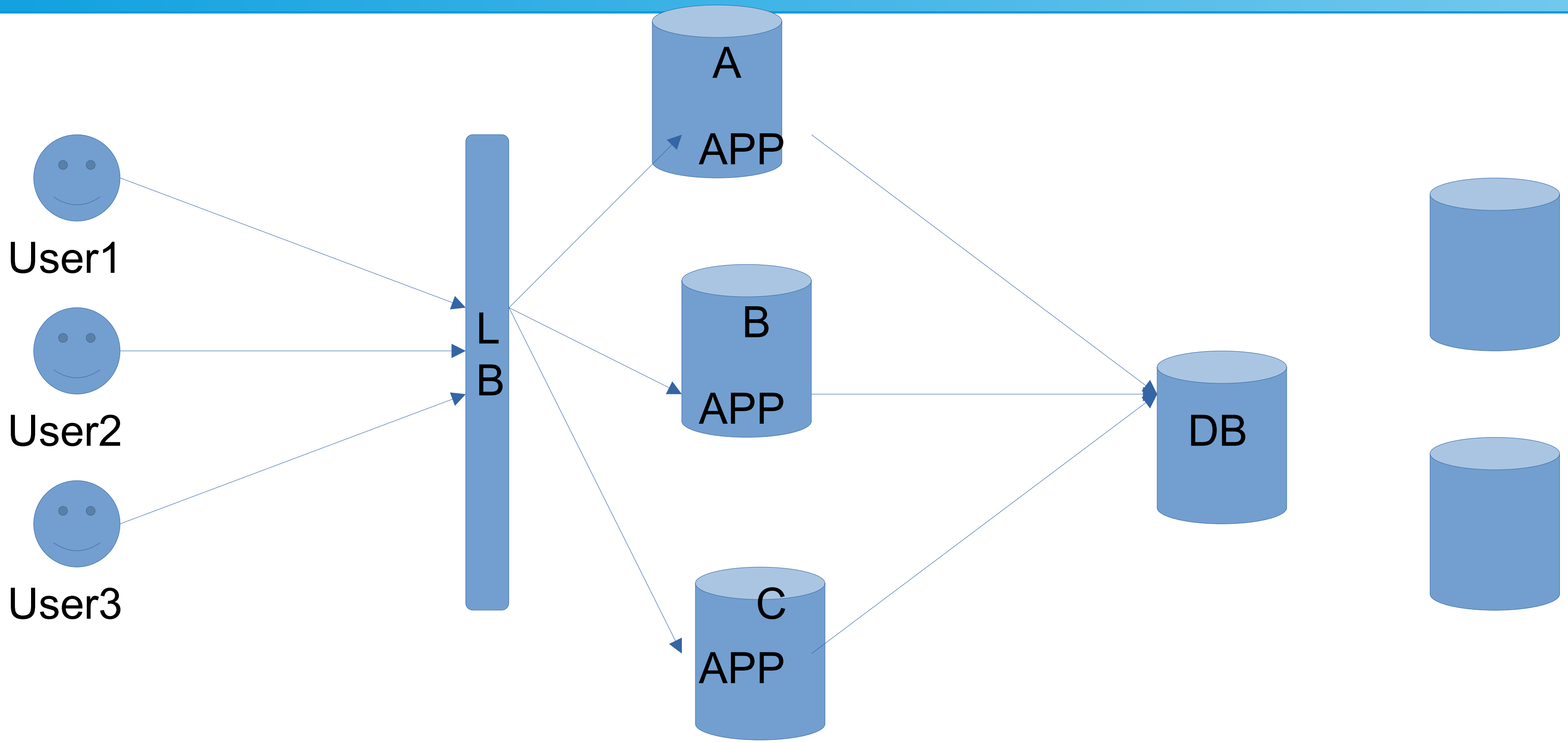
MySQL DBA – DAY 1

Terminologies

- Server
- Client
- IP address
- Port
- Primary
- Standby
- Load balancer
- APP server
- Web server
- Process

APP FLOW



RDBMS

- Software used to store, manage, query, and retrieve data stored in a relational database is called a relational database management system (RDBMS). The RDBMS provides an interface between users and applications and the database, as well as administrative functions for managing data storage, access, and performance.
- SQL – Standard query language

Limitations

- Maintenance
- Cost
- Physical storage
- Scalability
- Complexity
- Performance over time

NoSQL

- NoSQL databases (aka "not only SQL") are non-tabular databases and store data differently than relational tables. NoSQL databases come in a variety of types based on their data model. The main types are document, key-value, wide-column, and graph. They provide flexible schemas and scale easily with large amounts of data and high user loads.
- Features – Flexible, Scaling, Fast, Ease of use

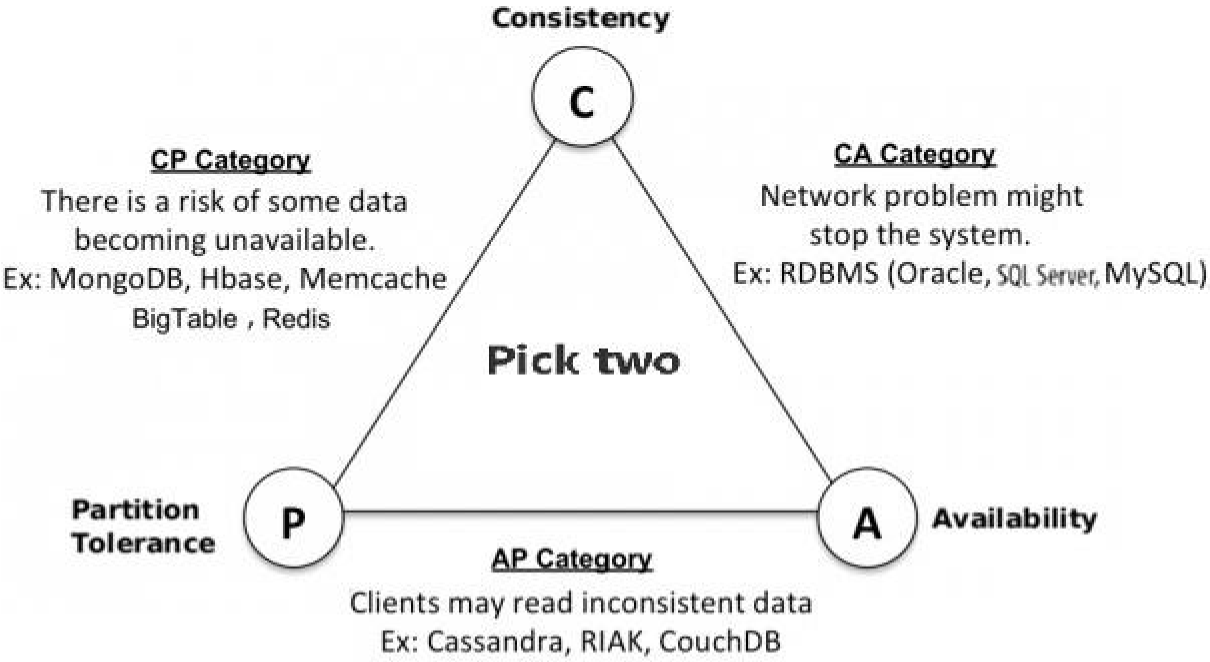
NoSQL

- Document databases store data in documents similar to JSON (JavaScript Object Notation) objects. Each document contains pairs of fields and values.
- Key-value databases are a simpler type of database where each item contains keys and values.
- Wide-column stores store data in tables, rows, and dynamic columns.
- Graph databases store data in nodes and edges. Nodes typically store information about people, places, and things, while edges store information about the relationships between the nodes.

RDBMS vs NoSQL

	SQL	NoSQL
Database Type	Relational Databases	Non-relational Databases / Distributed Databases
Structure	Table-based	• Key-value pairs • Document-based • Graph databases • Wide-column stores
Scalability	Designed for scaling up vertically by upgrading one expensive custom-built hardware	Designed for scaling out horizontally by using shards to distribute load across multiple commodity (inexpensive) hardware
Strength	• Great for highly structured data and don't anticipate changes to the database structure • Working with complex queries and reports	• Pairs well with fast paced, agile development teams • Data consistency and integrity is not top priority • Expecting high transaction load

CAP theorem


Consistency
C
CP Category
There is a risk of some data
becoming unavailable.
Ex: MongoDB, Hbase, Memcache
BigTable , Redis
CA Category
Network problem might
stop the system.
Ex: RDBMS (Oracle, SQL Server, MySQL)
Pick two
Partition
Tolerance
P
A
Availability
AP Category
Clients may read inconsistent data
Ex: Cassandra, RIAK, CouchDB

MySQL

- MySQL is popular among all databases, and is ranked as the 2nd most popular database, only slightly trailing Oracle Database. Among open source databases, MySQL is the most popular database in use today. Known as one of the most reliable and performative databases out there.
- MySQL is licensed under the GNU General Public License and is also available under several proprietary licenses. When Oracle bought MySQL AB in 2010, Michael "Monty" Widenius, MySQL founder, forked MySQL into a free, open source database called MariaDB -- with the intention of keeping the MariaDB project free and open source forever.
- MySQL has several versions available, but there are essentially two options: a community version, which is free to use; and paid versions, which include additional functionality, extensions, and support through Oracle. Despite the branding for the paid version, the community version is still considered to be production-ready and is often used in the enterprise.

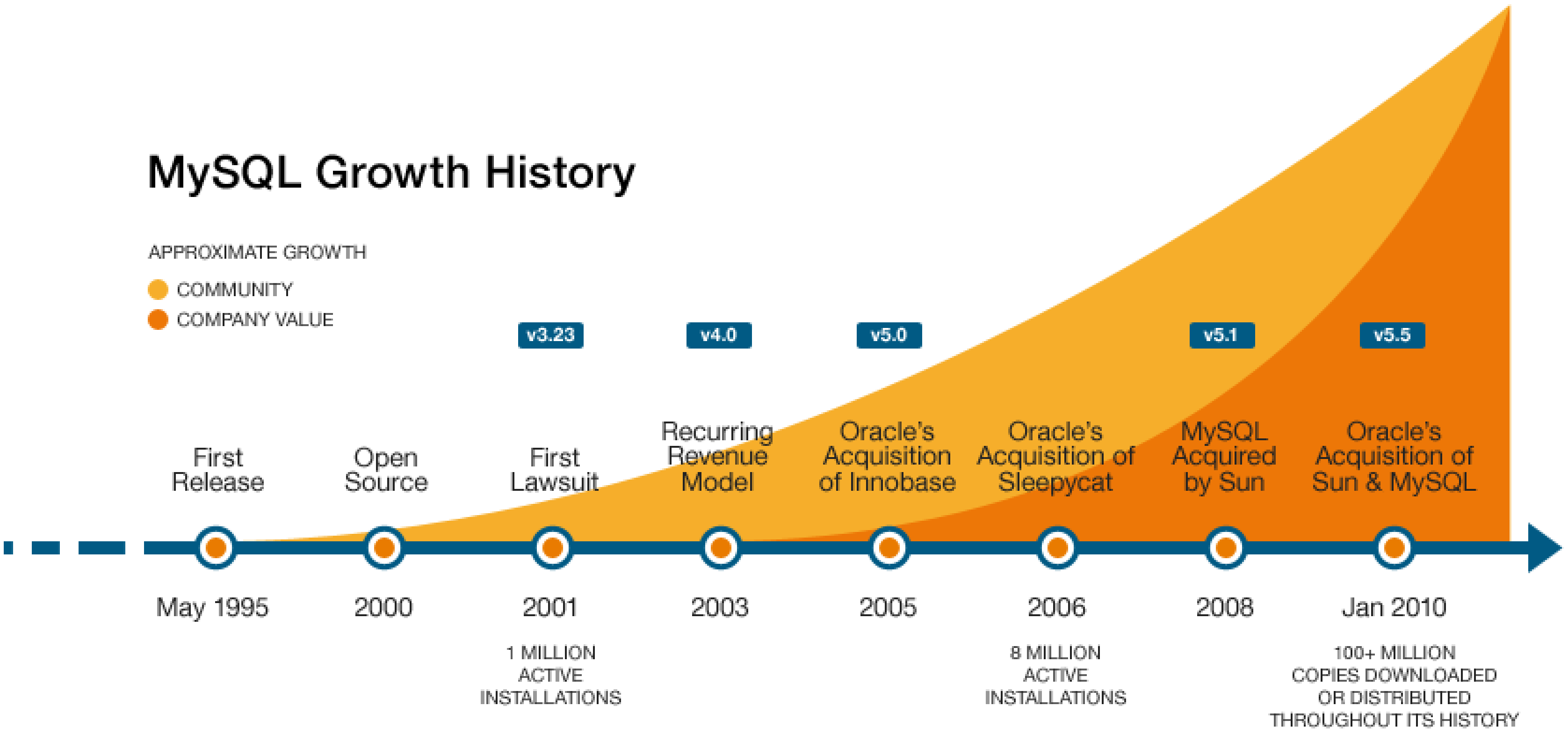
MySQL

- Easy to Use - MySQL is considered easy to use among RDBMS. It works with basic SQL and, given its maturity and adoption, there is abundant documentation available.
- Secure - MySQL's maturity also lends itself to security. It's regularly updated, has a vibrant developer community, and, because of it's wide adoption within the enterprise, many CVE patches are released before the CVE is announced. These factors combine to make MySQL a stable and secure choice among RDBMS.
- Open Source - The community edition of MySQL is enterprise ready, and supported by a GNU General Public License. For users who want access to equitable proprietary functionality of MySQL without the added price tag, there are other options within the ecosystem — like MariaDB — that can add similar levels of functionality and beyond.
- Scalable - MySQL is highly scalable for an RDBMS, with a wide range of options not covered in this blog that allow for tuning, customizing and enhancing your MySQL experience.
- Reliable - MySQL is reliable — not just from a data perspective, but from a development perspective. It's mature, it has regular releases, patches, and an entrenched developer community that works with it. This makes it a safe choice compared to newer, less mature RDBMS options.

MySQL timeline

Version	Year
V1	1995
V3	1996
V4	2002
V5	2005
V5.6	2013
V5.7	2015
V8	2018

MySQL history



SQL language

- DML – Data manipulation language – SELECT, INSERT, UPDATE, DELETE, LOCK,CALL, EXPLAIN PLAN
- DCL – Data control language – GRANT, REVOKE
- DDL – Data definition language - CREATE, DROP, ALTER, TRUNCATE, COMMENT, RENAME